

```
import pandas
from sklearn import linear_model

df = pandas.read_csv("cars.csv")

X = df[['Weight', 'Volume']]
y = df['CO2']

regr = linear_model.LinearRegression()
regr.fit(X, y)

#predict the CO2 emission of a car where the weight is 2300kg, and the volume is 1300cm3:
predictedCO2 = regr.predict([[2300, 1300]])

print(predictedCO2)
```

```
[107.2087328]
```

✓ Coefficient

The coefficient is a factor that describes the relationship with an unknown variable. In this case, we can ask for the coefficient value of weight against CO2, and for volume against CO2. The answer(s) we get tells us what would happen if we increase, or decrease, one of the independent values.

```
print(regr.coef_)
```

```
[0.00755095 0.00780526]
```

The result array represents the coefficient values of weight and volume.

Weight: 0.00755095 Volume: 0.00780526

These values tell us that if the weight increase by 1kg, the CO2 emission increases by 0.00755095g.

And if the engine size (Volume) increases by 1 cm3, the CO2 emission increases by 0.00780526 g.

I think that is a fair guess, but let test it!

We have already predicted that if a car with a 1300cm3 engine weighs 2300kg, the CO2 emission will be approximately 107g.

What if we increase the weight with 1000kg (from 2300 to 3300) what will be the CO2 emission?

Ans: $107.2087328 + (1000 * 0.00755095) = 114.75968$

```
predictedCO2 = regr.predict([[3300, 1300]])
```

```
print(predictedCO2)
```

```
[114.75968007]
```

✓ Amendments

```
import pandas as pd
from sklearn import linear_model

# Create a sample dataset
df = pd.DataFrame({
    "Weight": [1800, 2000, 2200, 2400, 2600, 2800, 3000, 3200],
    "Volume": [1000, 1200, 1300, 1400, 1600, 1800, 2000, 2200],
    "CO2": [95, 102, 108, 114, 122, 131, 140, 150]
})

X = df[['Weight', 'Volume']]
y = df['CO2']

regr = linear_model.LinearRegression()
regr.fit(X, y)

# Predict the CO2 emission for a car
predictedCO2 = regr.predict([[2300, 1300]])

print("Predicted CO2:", predictedCO2)
```

```
Predicted CO2: [109.07655039]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names: Feature names are not present.
warnings.warn(
```

```
predictedCO2 = regr.predict([[2500, 1700]])

print("New prediction:", predictedCO2)
```

```
New prediction: [125.20639535]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names: Feature names are not present.
warnings.warn(
```

```
new_car = pd.DataFrame({
    "Weight": [2500],
    "Volume": [1700]
})

predictedCO2 = regr.predict(new_car)

print("New prediction:", predictedCO2)
```

```
New prediction: [125.20639535]
```

✓ Conclusion

I created a sample dataset to demonstrate multiple linear regression as the original dataset (cars.csv) was not made available to me to download. The model was trained using vehicle weight and engine volume to predict CO2 emissions. After fitting the model, I tested it with a vehicle weighing 2500 kg with a 1700 cc engine. The model predicted that the CO2 emissions would approximately be 125.21, showing how multiple independent variables can be combined to estimate a dependent variable. I updated the prediction to use a pandas DataFrame with matching feature names to eliminate the scikit-learn warning and to follow recommended coding practices.

Start coding or [generate](#) with AI.